

```
In [17]: import pandas as pd
data = pd.read_csv("D:\\Data Engineering\\Python\\car_data_MV.csv")
import numpy as np
```

```
In [3]: data.shape
```

```
Out[3]: (301, 8)
```

```
In [4]: data.head()
```

```
Out[4]:
```

	Car_Name	Year	Selling_Price	Present_Price	Kms_Driven	Fuel_Type	Seller_Type	Transmission
0	ritz	2014.0	3.35	5.59	27000.0	Petrol	Dealer	Manual
1	sx4	NaN	4.75	NaN	43000.0	NaN	NaN	Manual
2	ciaz	NaN	7.25	NaN	6900.0	NaN	NaN	NaN
3	wagon r	NaN	2.85	NaN	5200.0	NaN	Dealer	NaN
4	swift	2014.0	4.60	6.87	42450.0	Diesel	Dealer	Manual

```
In [5]: data.info()      #to get dataframe info...
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 301 entries, 0 to 300
Data columns (total 8 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   Car_Name        298 non-null   object
1   Year            290 non-null   float64
2   Selling_Price   292 non-null   float64
3   Present_Price   286 non-null   float64
4   Kms_Driven      292 non-null   float64
5   Fuel_Type       294 non-null   object
6   Seller_Type     291 non-null   object
7   Transmission    291 non-null   object
dtypes: float64(4), object(4)
memory usage: 18.9+ KB
```

```
In [6]: data.isna()      #is it nulls?
```

```
Out[6]:
```

	Car_Name	Year	Selling_Price	Present_Price	Kms_Driven	Fuel_Type	Seller_Type	Transmission
0	False	False	False	False	False	False	False	False
1	False	True	False	True	False	True	True	False
2	False	True	False	True	False	True	True	True
3	False	True	False	True	False	True	False	True
4	False	False	False	False	False	False	False	False
...	...	...	...	...	...	...	...	...
296	False	False	False	False	False	False	False	False
297	False	False	False	False	False	False	False	False
298	False	False	False	False	False	False	False	False
299	False	False	False	False	False	False	False	False
300	False	False	False	False	False	False	False	False

301 rows x 8 columns

```
In [7]: data.isna().sum()      #counting nulls
```

```
Out[7]: Car_Name      3
Year            11
Selling_Price    9
Present_Price    15
Kms_Driven       9
Fuel_Type        7
Seller_Type     10
Transmission     10
dtype: int64
```

```
In [8]: data.shape
```

```
Out[8]: (301, 8)
```

```
In [10]: (data.isna().sum()/data.shape[0]) * 100 #getting the missing values percentage
```

```
Out[10]: Car_Name      0.996678
Year          3.654485
Selling_Price 2.990033
Present_Price 4.983389
Kms_Driven    2.990033
Fuel_Type     2.325581
Seller_Type   3.322259
Transmission  3.322259
dtype: float64
```

```
In [ ]: #impute missing values
```

```
In [ ]: #if missing values > 75% (simply drop that column)
#if missing values 30 to 75% (ask clients to get clarity)
#if missing values < 30% (impute it)
```

```
In [19]: data = {
    "Name" : ["Kiran", "Rupesh", "Sridhar", "Ranjita", "Neelu"],
    "Age"  : [54, None, 37, None, 41],
    "Salary" : [2300, 4500, None, 8000, 100000],
    "Department" : ["HR", "HR", "HR", np.NaN, "IT"]
}
```

```
In [20]: type(data)
```

```
Out[20]: dict
```

```
In [21]: df = pd.DataFrame(data)
```

```
In [22]: df
```

```
Out[22]:
```

	Name	Age	Salary	Department
0	Kiran	54.0	2300.0	HR
1	Rupesh	NaN	4500.0	HR
2	Sridhar	37.0	NaN	HR
3	Ranjita	NaN	8000.0	NaN
4	Neelu	41.0	100000.0	IT

```
In [23]: df.shape
```

```
Out[23]: (5, 4)
```

```
In [25]: df.isna()
```

```
Out[25]:
```

	Name	Age	Salary	Department
0	False	False	False	False
1	False	True	False	False
2	False	False	True	False
3	False	True	False	True
4	False	False	False	False

```
In [26]: df['Age'].fillna(df['Age'].mean(), inplace = True) # for directly replacing in column it self we can use inplace = True
```

```
In [28]: df['Age'].mean()
```

```
Out[28]: 44.0
```

```
In [ ]: df['Age'] = df['Age'].fillna(df['Age'].mean()) #we can assign it column
```

In [27]: df

Out[27]:

	Name	Age	Salary	Department
0	Kiran	54.0	2300.0	HR
1	Rupesh	44.0	4500.0	HR
2	Sridhar	37.0	NaN	HR
3	Ranjita	44.0	8000.0	NaN
4	Neelu	41.0	100000.0	IT

In []: df['Salary'].fillna(df['Salary'].median(), inplace = True)

In [29]: df['Salary'] = df['Salary'].fillna(df['Salary'].median())

In [30]: df

Out[30]:

	Name	Age	Salary	Department
0	Kiran	54.0	2300.0	HR
1	Rupesh	44.0	4500.0	HR
2	Sridhar	37.0	6250.0	HR
3	Ranjita	44.0	8000.0	NaN
4	Neelu	41.0	100000.0	IT

In [33]: df['Department'].mode()

Out[33]: 0 HR
Name: Department, dtype: object

In [37]: df['Department'] = df['Department'].fillna(df['Department'].mode()[0]) *#mode will fill you the most repeted values in blank*

In [38]: df

Out[38]:

	Name	Age	Salary	Department
0	Kiran	54.0	2300.0	HR
1	Rupesh	44.0	4500.0	HR
2	Sridhar	37.0	6250.0	HR
3	Ranjita	44.0	8000.0	HR
4	Neelu	41.0	100000.0	IT

In []: